

Mobile-First Options Decision Layer - Product and Engineering Plan

Project: BRstock1 Extension
Date: 2026-05-15
Status: Planning
Pain Origin: Real trading experience across Schwab, Moomoo, IBKR

1. The Pain Point — Why This Matters

1.1 What Every Options App Gets Wrong

All existing options platforms (Schwab, Moomoo, IBKR, Robinhood, Webull) are built from a **brokerage execution perspective**:

- How to place an order
- How to route fills
- How to display a full option chain

But the user’s real question is always:

“What should I do right now with this position?”

Specifically:

Real Decision Question	What Brokers Show Instead
Should I roll this spread?	Raw chain table with 200+ rows
What’s my new breakeven after rolling?	Nothing — mental math required
Is IV high or low right now?	A single IV number with no context
How much theta am I bleeding daily?	Greeks buried in a sub-menu
What’s my total portfolio delta risk?	Individual position Greeks only
Is this spread mainly vega or delta?	No dominant-risk indicator
What’s my recovery probability?	Not available anywhere

1.2 Why Mobile Makes It 10x Worse

Options are the **highest information-density instrument** in retail trading:

- Strike × Expiration × IV × Greeks × Skew × Term Structure × Assignment Risk × Theta Decay × Position Interaction

On desktop, this is barely manageable. On mobile:

- Tables are too dense to read
- Numbers are too small to parse
- Users must mentally cross-reference multiple screens
- Multi-leg spreads are completely non-intuitive
- Roll comparison requires 3-4 screen switches

Result: Users are “manually parsing risk” on a 6-inch screen. This is the gap.

1.3 The Opportunity

Build a **decision layer**, not a **trading execution layer**.

The product doesn’t compete with brokers on order routing. It competes on **compressed, actionable insight** — the thing brokers don’t provide:

"Your QQQ 700/720 bull call spread has 18 DTE remaining.
Current breakeven: 711.55. QQQ is at 708.
IV is at 78th percentile (past 1 year).
This position is now primarily theta-driven - you're losing \$18/day.
Rolling to June 720/740 would lower breakeven to 722 but costs \$3.20 net debit.
Recommendation: Hold if you expect QQQ above 712 in 18 days."

No broker gives you this. **BRstock1 can.**

2. BRstock1 Existing Asset Audit

Before building anything new, here’s what the project already has:

2.1 Direct Reuse — Ready Now

Module	Location	Reuse For
OptionSynth Engine	options_synth.py (919 lines)	Black-Scholes pricing, Greeks, IV estimation, chain generation
Schwab Market Data	schwab_market_data.py (444 lines)	Real-time option chains, quotes, normalized schema
Options Calibration	options_calibration.py (226 lines)	Synth-vs-real IV correction factors
Options Routes	routes/options.py (596 lines)	Schwab chain, synthetic chain, calibration APIs
Backtest Engine	options_synth.py — OptionBacktester	Wheels, LEAPS, vertical spreads backtesting
AI Analysis Pipeline	routes/stocks.py — Gemini/OpenRouter/Local	Structured AI report generation
FastAPI Backend	main.py + modular routes	API framework with auth, DB, static serving
Bilingual UI	app.js i18n system	EN/ZH language switching

2.2 Needs Extension

Capability	Current State	What’s Needed
Position Tracking	No real position storage	New: <code>user_option_positions</code> table
IV Percentile	HV_20d + RSI sentiment	New: Historical IV series for percentile ranking
Roll Analysis	Not implemented	New: Roll comparison calculator
Portfolio Greeks	Individual contract Greeks	New: Aggregate position-level Greeks
Mobile UI	Desktop-only static HTML	New: Responsive mobile-first pages
Push Insights	None	New: Decision alert engine

2.3 Key Advantage

BRstock1 already has what most startups spend months building:

1. **Working Black-Scholes + Greeks engine** with calibration
2. **Live Schwab data integration** with OAuth token management
3. **Synthetic chain fallback** when real data is unavailable
4. **AI report generation** with structured input
5. **Bilingual support** baked into the UI framework

The decision layer is an **integration and presentation problem**, not a computation problem.

3. Product Definition

3.1 Core Concept

Options Copilot - A mobile-first decision surface layered on top of broker data.

Not a broker. Not a charting app. A **decision compression engine** that answers:

1. **Position Health:** “How is my current position doing?”
2. **Action Clarity:** “Should I hold, roll, close, or adjust?”
3. **Risk Awareness:** “What’s my real exposure right now?”
4. **IV Context:** “Am I buying expensive or cheap volatility?”
5. **What-If Scenarios:** “If I roll to X, what changes?”

3.2 User Flow (Mobile-First)

```
+-----+
| [Mobile] Options Copilot |
|                             |
| +-----+                  |
| |   QQQ 700/720 BCS   |   |
| | (Yellow) 18 DTE | -$340 |
| | BE: 711.55 | Delta 0.32 |
| | Theta: -$18/day      |
| | IV Rank: 78%  ###.   |
| | [Roll] [Close] [AI]  |
| +-----+                  |
|                             |
| +-----+                  |
| |   TSLA LEAPS 180C   |   |
| | (Green) 245 DTE | +$2.1k |
| | Delta 0.78 | Theta: -$8/day |
| | IV Rank: 45%  ##..   |
| | [Details] [AI]       |
| +-----+                  |
|                             |
| -- Portfolio Risk --      |
| Net Delta: +0.65 | Net Theta: -$26 |
| Total Risk: $4,200        |
| Max Loss Today: -$890     |
+-----+
```

3.3 Decision Cards - The Core UI Primitive

Each position renders as a **Decision Card** that shows:

Layer	Content	Source
Header	Position name + structure	User input / broker import
Status	P/L, DTE, color-coded health	OptionSynth real-time calc
Key Metrics	Breakeven, dominant Greek, IV rank	OptionSynth + historical IV
Theta Burn	Daily/weekly dollar decay	OptionSynth Greeks
Risk Gauge	Primary risk factor (theta/vega/delta)	Greek analysis
Actions	Roll / Close / AI Analysis buttons	UI actions

3.4 AI Decision Reports

When user taps “AI” on a position card:

Input to AI (structured JSON):

```
{
  "position": "QQQ 700/720 Bull Call Spread",
  "current_price": 708.50,
  "entry_debit": 11.55,
  "current_value": 8.20,
  "breakeven": 711.55,
  "dte": 18,
  "greeks": {"delta": 0.32, "theta": -0.18, "vega": 0.35, "gamma": 0.008},
  "iv_rank_1y": 78,
  "hv_20": 0.22,
  "rsi": 48,
  "underlying_trend": "neutral-bearish",
  "roll_options": [
    {"target": "Jun 720/740", "net_cost": 3.20, "new_breakeven": 723.20, "new_dte": 48}
  ]
}
```

AI Output (compressed for mobile):

[Report] Position Assessment: HOLD with caution

Your QQQ 700/720 spread is -29% from entry with 18 DTE.
QQQ needs to reach 712 for breakeven - that's +0.5% from here.

[Warning] IV is elevated (78th percentile). This helps your long leg
but time decay is now dominant at \$18/day.

[Roll] Roll Analysis: Rolling to Jun 720/740 costs \$3.20 and extends
DTE by 30 days but raises breakeven from 712 -> 723.
Only attractive if you expect QQQ above 725.

[Tip] Suggestion: If QQQ doesn't recover above 710 within 5 days,
consider closing at ~30% loss rather than rolling into a
higher-breakeven position.

4. Technical Architecture

4.1 New Backend Modules

```
services/backend_api/
|-- routes/
|   |-- options.py           -- Existing (extend)
|   |-- positions.py        -- [New] Position management & decision APIs
|   |-- decision.py         -- [New] AI decision report generation
|-- options_synth.py        -- Existing (extend with roll calculator)
|-- position_analyzer.py    -- [New] Portfolio-level risk aggregation
+-- iv_history.py           -- [New] Historical IV percentile tracker
```

4.2 New Database Tables

```
-- User option positions (manual entry or broker import)
CREATE TABLE IF NOT EXISTS user_option_positions (
    id          SERIAL PRIMARY KEY,
    user_id     VARCHAR(100) NOT NULL,
    ticker      VARCHAR(10) NOT NULL,
    position_type VARCHAR(30) NOT NULL,      -- 'BULL_CALL_SPREAD', 'LEAPS', etc.
    legs        JSONB NOT NULL,              -- [{type, strike, expiry, qty, entry_price}]
    entry_date  DATE NOT NULL,
    entry_debit DECIMAL(12,2),
    status      VARCHAR(20) DEFAULT 'OPEN', -- OPEN / CLOSED / ROLLED
    notes       TEXT,
    created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Historical IV snapshots for percentile calculation
CREATE TABLE IF NOT EXISTS iv_history (
    id          SERIAL PRIMARY KEY,
    ticker      VARCHAR(10) NOT NULL,
    date        DATE NOT NULL,
    hv_20       DECIMAL(8,6),
    synth_atm_iv_30d DECIMAL(8,6),
    rsi         DECIMAL(6,2),
    spot_price  DECIMAL(12,2),
    UNIQUE(ticker, date)
);

-- Decision reports cache
CREATE TABLE IF NOT EXISTS option_decision_reports (
    id          SERIAL PRIMARY KEY,
    user_id     VARCHAR(100) NOT NULL,
```

```

position_id INTEGER REFERENCES user_option_positions(id),
report_type VARCHAR(30),
language    VARCHAR(5) DEFAULT 'en',
report      TEXT NOT NULL,
input_snapshot JSONB,
created_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

4.3 New API Endpoints

Position Management

```

POST  /api/positions          - Add a position (manual entry)
GET    /api/positions          - List user's open positions
GET    /api/positions/{id}/snapshot - Live snapshot: Greeks, P/L, IV rank
PUT    /api/positions/{id}     - Update position (roll, close, notes)
DELETE /api/positions/{id}     - Remove position

```

Decision Engine

```

GET    /api/positions/{id}/decision - AI decision report for one position
GET    /api/positions/portfolio-risk - Aggregate portfolio Greeks & risk
GET    /api/positions/{id}/roll-analysis?target_dte=48&target_width=20

```

IV Context

```

GET    /api/iv/{ticker}/percentile - Current IV rank (1Y, 6M, 3M)
GET    /api/iv/{ticker}/history    - Historical IV series for charting

```

4.4 Position Analyzer Core Logic

```

class PositionAnalyzer:
    """Aggregates real-time analytics for user option positions."""

    def snapshot(self, position, current_price, chain_or_synth):
        """Returns a decision-ready snapshot for one position."""
        return {
            "current_value": ...,
            "unrealized_pl": ...,
            "breakeven": ...,
            "dte": ...,
            "net_delta": ...,
            "net_theta": ...,      # Daily dollar theta
            "net_vega": ...,
            "dominant_greek": ..., # "theta" / "vega" / "delta"
            "iv_rank_1y": ...,
            "health": ...,        # "healthy" / "at_risk" / "critical"
        }

    def portfolio_risk(self, positions, current_prices):
        """Aggregate risk across all open positions."""
        return {
            "total_net_delta": ...,
            "total_net_theta_daily": ...,
            "total_risk_at_work": ...,
            "max_loss_today": ...,
            "concentration_warning": ...,
        }

    def roll_analysis(self, position, current_price, target_dte, target_width):
        """Compare current position vs proposed roll."""
        return {
            "current": { "breakeven": ..., "max_profit": ..., "dte": ... },
            "rolled": { "breakeven": ..., "max_profit": ..., "net_cost": ... },
            "cost_per_day_gained": ...,
            "recommendation": ...,
        }

```

}

5. Mobile UI Architecture

5.1 Approach: Responsive Web, Not Native App

Build as a responsive web page within BRstock1, not a separate native app.

Rationale: - BRstock1 already serves static HTML via FastAPI - No app store approval needed - Users can “Add to Home Screen” for app-like experience - Reuses existing auth, API, and i18n infrastructure

5.2 Mobile Design Principles

1. **Card-based layout** - One position = one swipeable card
 2. **Color-coded health** - Green/Yellow/Red instant status
 3. **Large touch targets** - Minimum 44px tap zones
 4. **Progressive disclosure** - Summary first, tap for details
 5. **Bottom-sheet actions** - Roll/Close/AI as bottom drawers
 6. **Dark theme default** - Matches trading environment
-

6. Implementation Roadmap

Phase 0: IV History Foundation (1 session)

Goal: Start collecting IV history — needed for percentile rankings.

- ☐ Create `iv_history` table
- ☐ Add daily IV snapshot job (runs with existing after-close scheduler)
- ☐ Backfill 1 year of synthetic ATM IV for QQQ, TSLA, AAPL, NVDA, SPY
- ☐ Add `GET /api/iv/{ticker}/percentile` endpoint

Why first: IV percentile is the single most valuable context signal. Every day without collection = one less data point.

Phase 1: Position Management Backend (1-2 sessions)

Goal: Users can manually enter and track option positions.

- ☐ Create `user_option_positions` table
- ☐ Create `routes/positions.py` with CRUD endpoints
- ☐ Implement `PositionAnalyzer.snapshot()` - live P/L, Greeks, breakeven
- ☐ Wire to existing OptionSynth for real-time pricing
- ☐ Wire to Schwab for live chain data when available

Phase 2: Decision Cards - Mobile UI MVP (2-3 sessions)

Goal: Working mobile-first position dashboard.

- ☐ Add `page-copilot` to `index.html` with responsive layout
- ☐ Build Decision Card component with health indicators
- ☐ Add position entry form (bottom sheet on mobile)
- ☐ Portfolio risk summary bar
- ☐ Theta burn counter (daily \$ loss across all positions)
- ☐ Test on iPhone Safari, Android Chrome

Phase 3: Roll Calculator (1 session)

Goal: Compare roll scenarios directly on the position card.

- ☐ Implement `PositionAnalyzer.roll_analysis()`
- ☐ Add roll comparison UI as expandable card section
- ☐ Show side-by-side: current vs rolled breakeven, cost, DTE
- ☐ Cost-per-day-gained metric

Phase 4: AI Decision Reports (1-2 sessions)

Goal: One-tap AI analysis for any position.

- ☐ Create `routes/decision.py` with structured AI prompt
- ☐ Reuse existing Gemini/OpenRouter pipeline
- ☐ Create `option_decision_reports` table for caching
- ☐ Mobile-optimized report display
- ☐ Bilingual report generation (EN/ZH)

Phase 5: Broker Position Import (future)

- ☐ Schwab positions API (requires trading scope)
- ☐ IBKR / Moomoo position sync (when available)
- ☐ Auto-detect position structure

7. Competitive Positioning

Feature	Brokers	Analytics Sites	BRstock1 Copilot
Mobile-first design	[NO] Desktop-first	[NO] Desktop-only	[YES] Built for phone
Decision compression	[NO] Raw data dump	[WARN] Charts only	[YES] Actionable cards
AI position analysis	[NO]	[NO]	[YES] Per-position AI
Roll comparison	[WARN] Manual only	[NO]	[YES] One-tap compare
IV percentile context	[WARN] Some	[YES]	[YES] On every card
Portfolio risk aggregate	[WARN] Basic	[NO]	[YES] Net Greeks + risk
Bilingual EN/ZH	[NO]	[NO]	[YES] Built-in
Synthetic chain fallback	[NO]	[NO]	[YES] Works offline

Wedge Strategy: Personal tool first -> if it saves 10+ min per decision and prevents 1-2 bad rolls/month -> real product value for any mobile options trader.

8. Open Questions

- Which broker will you use most? Determines position import priority.
- How many active positions do you typically hold? Affects portfolio view design.
- Real-time updates or 30-second refresh acceptable?
- Separate page or replace existing Options Analysis page?
- Start Phase 0 (IV history) immediately or Phase 1 (positions)?

9. Summary

BRstock1 is uniquely positioned to build this:

- [YES] **Math engine** exists (OptionSynth + Greeks + B-S + calibration)
- [YES] **Data pipeline** exists (Schwab + synthetic fallback)
- [YES] **AI infrastructure** exists (Gemini + OpenRouter + bilingual)
- [YES] **Backend framework** exists (FastAPI + auth + DB)
- [YES] **You are the target user** - every design decision comes from real pain

What’s missing is the **presentation layer** that compresses all this into mobile-friendly decision cards. That’s a UI/UX problem, not an infrastructure problem.

Recommended first action: Start Phase 0 today — begin collecting IV history data. It costs almost nothing and creates irreplaceable time-series data.